

COM3105 Advanced Algorithms

Homework 4

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Question 1

(5 points)

Consider a stream $a_1, a_2, \dots, a_m$, where each $a_i \in [n]$.

Design and analyze a streaming algorithm that, given a query

$S \subseteq [n]$,

outputs an estimate of what fraction of items in the stream belong to S within additive error ϵ with probability at least δ .

The space used by the algorithm should only depend on ϵ and δ .

Note that S is given only at query time (not in advance).

Answer

The intuition behind my answer is to use reservoir sampling with k samples to estimate the fraction of streamed elements in S .

Algorithm:

- **Initialisation:** Choose a sample size k based on the parameters:

$$k \geq \frac{1}{4\epsilon^2\delta}$$

Where ϵ is the chosen additive error and δ is the probability of failure. Initialise an empty set (the reservoir) R that stores k items. Initialise a counter $C = 0$ to track the number of items seen so far.

- **Processing the stream:** For each incoming element a_i , where i is the index of the element in the stream (starting from 1):
 - If $C < k$, insert a_i into R , and increment C by 1.
 - Otherwise:
 - ◊ Increment C by 1.
 - ◊ Include a_i in the reservoir R with probability k/C .
 - ◊ If included, replace a uniformly random element from R .

This is standard reservoir sampling.

- **Query:** When the set S is given, compute the estimate:

$$\hat{p} = \frac{1}{k} \sum_{x \in R} \mathbf{1}\{x \in S\},$$

and output $\hat{p}$ as the estimate of the true fraction.

Expectation of the Estimator:

Let p be the true fraction of stream elements that belong to S :

$$p = \frac{1}{m} |\{i : a_i \in S\}|$$

After reservoir sampling has finished, the reservoir contains k uniformly random elements from the stream. For each $j = 1, \dots, k$, define the indicator variable

$$X_j = \mathbf{1}\{R_j \in S\},$$

where R_j is the j -th sampled item.

Because reservoir sampling picks each stream element with equal probability, each R_j is distributed exactly as a uniformly random stream element. So:

$$\mathbb{P}(R_j \in S) = p,$$

and therefore:

$$\mathbb{E}[X_j] = 1 \cdot p + 0 \cdot (1 - p) = p$$

The estimator for the fraction is:

$$\hat{p} = \frac{1}{k} \sum_{j=1}^k X_j$$

Using linearity of expectation:

$$\begin{aligned} \mathbb{E}[\hat{p}] &= \mathbb{E}\left[\frac{1}{k} \sum_{j=1}^k X_j\right] \\ &= \frac{1}{k} \sum_{j=1}^k \mathbb{E}[X_j] \\ &= \frac{1}{k} \sum_{j=1}^k p \\ &= \frac{1}{k} \cdot kp \\ &= p \end{aligned}$$

So the estimator has expectation equal to the true fraction:

$$\mathbb{E}[\hat{p}] = p$$

Variance of the Estimator:

Remember that each random variable X_j is an indicator:

$$X_j = \begin{cases} 1 & \text{if the } j\text{-th sampled element belongs to } S, \\ 0 & \text{otherwise} \end{cases}$$

So,

$$\mathbb{P}(X_j = 1) = p \quad \text{and} \quad \mathbb{P}(X_j = 0) = 1 - p$$

To get the variance of X_j we can use the formula:

$$\text{Var}(X_j) = \mathbb{E}[X_j^2] - (\mathbb{E}[X_j])^2$$

Since $X_j^2 = X_j$ (as the indicators are either 0 or 1):

$$\mathbb{E}[X_j^2] = \mathbb{E}[X_j]$$

And from the expectation calculation above we have:

$$\mathbb{E}[X_j] = p$$

Putting this together:

$$\text{Var}(X_j) = p - p^2 = p(1 - p).$$

Because $p(1 - p)$ is maximized when $p = \frac{1}{2}$, we know that:

$$\text{Var}(X_j) = p(1 - p) \leq \frac{1}{4}$$

Remembering that the estimator is the mean of the X_j in the reservoir:

$$\hat{p} = \frac{1}{k} \sum_{j=1}^k X_j$$

Remembering the two properties of variance:

- For any random variable Y and constant c ,

$$\text{Var}(cY) = c^2 \text{Var}(Y)$$

- For independent random variables Y_1, Y_2 ,

$$\text{Var}(Y_1 + Y_2) = \text{Var}(Y_1) + \text{Var}(Y_2)$$

As the X_j are independent, we can use these properties to compute the variance of $\hat{p}$:

$$\text{Var}(\hat{p}) = \text{Var}\left(\frac{1}{k} \sum_{j=1}^k X_j\right) = \frac{1}{k^2} \sum_{j=1}^k \text{Var}(X_j)$$

Using $\text{Var}(X_j) \leq 1/4$ for every j :

$$\text{Var}(\hat{p}) \leq \frac{1}{k^2} \cdot k \cdot \frac{1}{4} = \frac{1}{4k}$$

Quality Analysis Using Chebyshev's Inequality:

Chebyshev's inequality is defined as:

$$\mathbb{P}(|X - \mu| > \lambda) \leq \frac{\text{Var}(X)}{\lambda^2}$$

Using Chebyshev's inequality for our estimator $\hat{p}$, we can substitute in our bounded variance (from above) and set $\lambda = \epsilon$:

$$\mathbb{P}(|\hat{p} - p| > \epsilon) \leq \frac{1}{4k\epsilon^2}$$

To bound the probability of error, we set the right-hand side to be at most δ' :

$$\frac{1}{4k\epsilon^2} \leq \delta'$$

Now we solve for k :

$$\frac{1}{4\epsilon^2\delta'} \leq k$$

Substituting this into the right-hand side:

$$\mathbb{P}(|\hat{p} - p| > \epsilon) \leq \frac{1}{4 \cdot \left(\frac{1}{4\epsilon^2\delta'}\right) \cdot \epsilon^2} = \frac{1}{\delta'} = \delta'$$

Therefore, with this choice of k we have:

$$\mathbb{P}(|\hat{p} - p| > \epsilon) \leq \delta'$$

Which we can rewrite as:

$$\mathbb{P}(|\hat{p} - p| \leq \epsilon) \geq 1 - \delta'$$

Now we can set $1 - \delta' = \delta$ to bound the probability of error by δ :

$$\mathbb{P}(|\hat{p} - p| \leq \epsilon) \geq \delta$$

This proves that the estimator $\hat{p}$ is within additive error ϵ of the true fraction p with probability at least δ .

Space Complexity Analysis:

The algorithm stores:

- A reservoir of size $k = \mathcal{O}\left(\frac{1}{4\epsilon^2(1-\delta)}\right)$.
- Each stored item in k is from $[n]$ and requires $\mathcal{O}(\log n)$ bits.
- The counter requires $\mathcal{O}(\log m)$ bits, where m is the length of the stream in the question.

Therefore, the total space used by the algorithm is:

$$\mathcal{O}(k \log n + \log m) = \mathcal{O}\left(\frac{1}{\epsilon^2(1-\delta)} \log n + \log m\right)$$

This space depends only on ϵ and δ (and $\log n$ and $\log m$, but these are not tuneable), as required.

Question 2a

(2 points)

An independent set in a graph is a set of vertices which does not contain any edge. The Maximum Independent Set (MIS) problem is the problem of finding an independent set of maximum size in a given graph.

Write down an integer program which solves MIS exactly. (2 points)

Answer

Let $G = (V, E)$, where V is the set of vertices and E is the set of edges of the graph.

For each vertex $v \in V$, define a binary variable x_v such that:

$$x_v = \begin{cases} 1 & \text{if vertex } v \text{ is included in the independent set,} \\ 0 & \text{otherwise} \end{cases}$$

Constraints:

$$\forall v \in V : \quad x_v \in \{0, 1\}$$

$$\forall (u, v) \in E : \quad x_u + x_v \leq 1$$

Objective Function:

$$\text{Maximize } \sum_{v \in V} x_v$$

Question 2b

(3 points)

An independent set in a graph is a set of vertices which does not contain any edge. The Maximum Independent Set (MIS) problem is the problem of finding an independent set of maximum size in a given graph.

Use the rounding technique to give a $(2 - \frac{n}{\alpha})$ -approximation algorithm for MIS, where n is the number of vertices and α is the size of the largest independent set in a given graph.

Answer

This is a relaxation-LP based approximation algorithm.

We define the same graph $G = (V, E)$, where V is the set of vertices and E is the set of edges of the graph.

Except that now the variables x_v are allowed to take any value in the range $[0, 1]$.

Constraints:

$$\forall v \in V : \quad 0 \leq x_v \leq 1$$

$$\forall (u, v) \in E : \quad x_u + x_v \leq 1$$

Objective Function:

$$\text{Maximize } \sum_{v \in V} x_v$$

Algorithm:

Let x_v^* be the optimal solution for each vertex $v \in V$ to the relaxed LP.

For each vertex $v \in V$:

$$\text{include } v \text{ in the independent set } I \text{ if } x_v^* > \frac{1}{2}$$

Return the independent set I .

Because of the constraints, no two adjacent vertices can both have $x_v^* > \frac{1}{2}$, so I is an independent set.

Approximation Analysis:

Let our optimal relaxation LP solution be $OPT_{RLP} = \sum_{v \in V} x_v^*$.

split the vertexes according to their value x_v^* :

$$H = \{v \in V : x_v^* > \frac{1}{2}\}$$

$$L = \{v \in V : x_v^* \leq \frac{1}{2}\}$$

Using this split we can simplify OPT_{RLP} :

$$OPT_{RLP} = \sum_{v \in H} x_v^* + \sum_{v \in L} x_v^*$$

In the set H , each $x_v^* \leq 1$, so each vertex contributes at most 1 to the sum:

$$\sum_{v \in H} x_v^* \leq |H|$$

Similarly, in the set L , each $x_v^* \leq \frac{1}{2}$, so each vertex contributes at most $\frac{1}{2}$ to the sum:

$$\sum_{v \in L} x_v^* \leq \frac{1}{2}|L|$$

Using these bounds for H and L we can bound OPT_{RLP} :

$$OPT_{RLP} \leq |H| + \frac{1}{2}|L|$$

As $|H| + |L| = n$, we can substitute $|L| = n - |H|$:

$$OPT_{RLP} \leq |H| + \frac{1}{2}(n - |H|)$$

$$OPT_{RLP} \leq |H| + \frac{n}{2} - \frac{1}{2}|H|$$

$$OPT_{RLP} \leq \frac{n}{2} + \frac{1}{2}|H|$$

We know that the true best solution is at most the optimal relaxed solution:

$$\alpha \leq OPT_{RLP}$$

Substituting our bound for OPT_{RLP} :

$$\alpha \leq \frac{n}{2} + \frac{1}{2}|H|$$

We can rearrange this to bound $|H|$:

$$2\alpha - n \leq |H|$$

In our algorithm, we choose all vertices in H to be in our independent set I , so:

$$|I| = |H|$$

Thus, we have:

$$|I| \geq 2\alpha - n$$

Therefore, the approximation ratio is:

$$\begin{aligned} \frac{|I|}{\alpha} &\geq \frac{2\alpha - n}{\alpha} \\ \frac{|I|}{\alpha} &\geq \frac{\alpha(2 - \frac{n}{\alpha})}{\alpha} \\ \frac{|I|}{\alpha} &\geq 2 - \frac{n}{\alpha} \end{aligned}$$

This shows that our relaxed LP algorithm is a $(2 - \frac{n}{\alpha})$ -approximation for the MIS problem.
